



FULHAM REACH BOAT CLUB

EST. 2014

Makeover Monday – River Water Quality

Problem Statement

Can we use weather, river flow and water quality data to better understand the forces that impact water in London at Fulham Reach Boat Club, located at Hammersmith Bridge.

Future Goal

With better understanding can we get better at predicting when water quality will be poor, this would be advantageous as we only have capacity to test once a week and the results of E.Coli testing takes a day to process.

Background and FRBC Open Data sets

Fulham Reach Boat Club (FRBC) is a sport for good charity that takes a holistic approach to health and wellbeing of participants and the environment we benefit from.

We collect data on the water quality as well as the amount of rubbish we collect during river clean ups.

FRBC has been testing water quality since August 2023 with the aim to understand the quality of the river environment.

We collect a range of data in a citizen science style, these are:

- physiochemical measures,
- E.coli (bacteria)
- and observational data such as weather, people involved, and tide (Edd – out or Flood - in)

This all helps us to build better understanding of the quality of the river.



Our water quality data set

file: FRBC FreshWater Watch.xlsx

Our river clean up data set

file: FRBC River Clean ups.xlsx

Interesting Case Study dates

On 20th Feb we recorded 563 Coliform Units (CFU) per 100 ml for E.Coli from the FRBC pontoon just by Hammersmith Bridge. This is much lower than what we have seen before, which would usually be over 1,000 CFU/100ml.

Conversely on 27th November we recorded 110,000 CFRU/100ml by far the highest reading we ever recorded, this followed a yellow weather warning storm.

Relevant Public Open Data sets

We are also aware of other data sources that would have correlations with the data we are collecting, or could be useful for understanding and predicting water quality:

- **Defra Hydrology Data**

Dissolved Oxygen data is another indicator of organic pollution as when organic pollution is high the dissolved oxygen will fall. Rainfall data and River flow rate data can also be indicators when the sewer system might be getting close to overloading and a Combined Sewer Overflow about to spill into the river.

<https://environment.data.gov.uk/hydrology/explore>

- **Met Office Rainfall Data**

Another source for rainfall data, as per above

<https://www.metoffice.gov.uk/research/approach/collaboration/ukcp/data/index>

- **Port of London Authority Flag Status**

A practical user flag warning system based on the water height on a low tide at Richmond.

<https://pla.co.uk/ebb-tide-flag-warning>

- How the flag colour is determined

Every day at 6 am and 6 pm the lowest tide reading for the Richmond tide gauge is taken for the preceding 12 hours.

If the value is 2.6 metres or above then the flag is RED.

If the value is 1.7 metres or above but less than 2.6 metres then the flag is YELLOW.

If the value is 0 or above but less than 1.7 metres then the flag is GREEN.

If the value is less than 0 then the flag is BLACK.

- **Combined Sewer Overflow data**

Monitors on Combined Sewer Overflows (CSOs) that track the time discharging, but not volumes of sewage.

<https://www.sewagemap.co.uk/>

<https://www.streamwaterdata.co.uk/pages/the-national-storm-overflow-hub>

- **The Tideway Tunnel / Super Sewer**

Captures CSOs Acton Storm overflow and Hammersmith CSO from now onwards, these are two heavily polluting CSOs.

<https://www.tideway.london/>